AFTER BLOWING UP AN ALGEBRAIC SPACE ALONG A FINITE TYPE IDEAL, A RATIONAL POINT OF A SEPARATED GROUP SPACE EXTENDS TO A SECTION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9024-9037 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 9039-9050, the complete author proof (10 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Spaces-side companion to the schemes get-section family collected in previous waves (L16057 slice-once etc.); paired check: the schemes chapter's analogue has a different statement shape (group schemes, not spaces) so counted separately per the paired rule with the category/structure difference noted.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Blow-ups of algebraic spaces along finite type ideals (their definition and universal property, chapter section), schematic closure of subsheaves (cited), the chapter's preceding flattening/extension lemmas, separatedness of the group space for uniqueness of extensions.

Source reference. Original lemma 087G: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9004–9012. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 086D: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9050–9050. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CP3: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9050–9050. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let X, Y be algebraic spaces over S. Let $U \subset W \subset Y$ be open subspaces. Let $f : X \to W$ and let $s : U \to X$ be morphisms such that $f \circ s = id_U$. Assume*

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

(1) *f is proper,*
(2) *Y is quasi-compact and quasi-separated, and*
(3) *U and W are quasi-compact.*

Then there exists a U-admissible blowup $b : Y' \to Y$ and a morphism $s' : b^{-1}(W) \to X$ extending s with $f \circ s' = b|_{b^{-1}(W)}$.

Proof. We may and do replace X by the scheme theoretic image of s. Then $X \to W$ is an isomorphism over U, see Morphisms of Spaces, Lemma 0CP3. By Lemma 087G there exists a U-admissible blowup $W' \to W$ and an extension $W' \to X$ of s. We finish the proof by applying Divisors on Spaces, Lemma 086D to extend $W' \to W$ to a U-admissible blowup of Y. □